

SOLIDWORKS API (.NET VBA C#) — Reference

(188/337): IModelDoc2 (part 11 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 11 of 17) (Type)

SOLIDWORKS API Help

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InsertExtendSurface Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertExtendSurface Method (IModelDoc2)

Extends a surface along the selected faces or edges.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertExtendSurface( _  
    ByVal ExtendLinear As System.Boolean, _  
    ByVal EndCondition As System.Integer, _  
    ByVal Distance As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ExtendLinear As System.Boolean  
Dim EndCondition As System.Integer  
Dim Distance As System.Double  
  
instance.InsertExtendSurface(ExtendLinear, EndCondition, Distance)
```

C#

```
void InsertExtendSurface(  
    System.bool ExtendLinear,  
    System.int EndCondition,  
    System.double Distance  
)
```

C++/CLI

```
void InsertExtendSurface(  
& System.bool ExtendLinear,  
& System.int EndCondition,  
& System.double Distance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ExtendLinear

True to extend surface linearly, false to extend along the same surface

EndCondition

- 0 - Extend surface by given distance
- 1 - Extend surface up to a selected point
- 2 - Extend surface up to a selected surface

Distance

Distance to extend surface along

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertExtendSurface](#).

Remarks

The selection list can contain faces or edges from the surface. These selected entities will be extended away from the surface according to the input arguments.

The selected point or surface to which to extend should be in the selection list. If EndCondition is to a selected surface, then currently only faces from solids are supported through the API.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISurfaceExtendFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertFamilyTableEdit Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertFamilyTableEdit Method (IModelDoc2)

Edits an open design table from Microsoft Excel.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertFamilyTableEdit()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertFamilyTableEdit()
```

C#

```
void InsertFamilyTableEdit()
```

C++/CLI

```
void InsertFamilyTableEdit();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertFamilyTableEdit](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::DeleteDesignTable Method](#)
[IModelDoc2::GetDesignTable Method](#)
[IModelDoc2::IGetDesignTable Method](#)
[IModelDoc2::InsertFamilyTableNew Method](#)
[IModelDoc2::InsertFamilyTableOpen Method](#)
[IModelDoc2::CloseFamilyTable Method](#)
[IModelDocExtension::HasDesignTable Method](#)
[IDesignTable Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertFamilyTableNew Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertFamilyTableNew Method (IModelDoc2)

Inserts an existing design table from the model into the selected drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertFamilyTableNew()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertFamilyTableNew()
```

C#

```
void InsertFamilyTableNew()
```

C++/CLI

```
void InsertFamilyTableNew();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertFamilyTableNew](#).

Remarks

Before using this method in a drawing, you must have selected a drawing view and that drawing view must contain a model that has an existing family table. If you have not selected a drawing view before using this method or if the selected drawing view does not contain a family table, then this method takes no action.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CloseFamilyTable Method](#)
[IModelDoc2::DeleteDesignTable Method](#)
[IModelDoc2::GetDesignTable Method](#)
[IModelDoc2::IGetDesignTable Method](#)
[IModelDoc2::InsertFamilyTableEdit Method](#)
[IModelDoc2::InsertFamilyTableOpen Method](#)
[IModelDocExtension::HasDesignTable Method](#)
[IDesignTable Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertFamilyTableOpen Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertFamilyTableOpen Method (IModelDoc2)

Inserts the specified Microsoft Excel design table.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFamilyTableOpen( _  
    ByVal FileName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim FileName As System.String
Dim value As System.Boolean

value = instance.InsertFamilyTableOpen(FileName)
```

C#

```
System.bool InsertFamilyTableOpen(
    System.string FileName
)
```

C++/CLI

```
System.bool InsertFamilyTableOpen(
    & System.String^ FileName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Full path and filename of the Microsoft Excel design table

Return Value

True if the design table is inserted, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertFamilyTableOpen](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CloseFamilyTable Method](#)
[IModelDoc2::DeleteDesignTable Method](#)
[IModelDoc2::GetDesignTable Method](#)
[IModelDoc2::IGetDesignTable Method](#)
[IModelDoc2::InsertFamilyTableEdit Method](#)
[IModelDoc2::InsertFamilyTableNew Method](#)
[IModelDocExtension::HasDesignTable Method](#)
[IDesignTable Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertFeatureReplaceFace Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertFeatureReplaceFace Method (IModelDoc2)

Creates a Replace Face feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertFeatureReplaceFace()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2

instance.InsertFeatureReplaceFace()
```

C#

```
void InsertFeatureReplaceFace()
```

C++/CLI

```
void InsertFeatureReplaceFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the edges match, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertFeatureReplaceFace](#).

Example

[Create Replace Face Feature \(VBA\)](#)
[Create Replace Face Feature \(VB.NET\)](#)
[Create Replace Face Feature \(C#\)](#)

Remarks

Before calling this method, select the faces to replace by calling [IModelDocExtension::SelectByID2](#) with Marks of 1 and select the replacement surfaces by calling IModelDocExtension::SelectByID2 with Marks of 2.

See SOLIDWORKS Help for details about the Replace Face feature.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IReplaceFaceFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertFeatureShell Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertFeatureShell Method (IModelDoc2)

Creates a shell feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertFeatureShell( _  
    ByVal Thickness As System.Double, _  
    ByVal Outward As System.Boolean _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Thickness As System.Double  
Dim Outward As System.Boolean
```

```
instance.InsertFeatureShell(Thickness, Outward)
```

C#

```
void InsertFeatureShell(  
    System.double Thickness,
```



```
        System.bool Outward
    )
```

C++/CLI

```
void InsertFeatureShell(
&    System.double Thickness,
&    System.bool Outward
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness
Shell thickness in meters

Outward
True for outside, false for inside

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertFeatureShell](#).

Example

[Create Shell Feature \(VBA\)](#)
[Create Shell Feature \(VB.NET\)](#)
[Create Shell Feature \(C#\)](#)

Remarks

Before calling this method, select faces to remove to create the shell using [IModelDocExtension::SelectByID2](#) with Marks of 1. See the SOLIDWORKS Help for information about what entities are valid for selection.

To make a multi-thickness shell, use this method with [IModelDoc2::InsertFeatureShellAddThickness](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IShellFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertFeatureShellAddThickness Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertFeatureShellAddThickness Method (IModelDoc2)

Adds thickness to a face in a multi-thickness shell feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertFeatureShellAddThickness( _
    ByVal Thickness As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Thickness As System.Double

instance.InsertFeatureShellAddThickness(Thickness)
```

C#

```
void InsertFeatureShellAddThickness(  
    System.double Thickness  
)
```

C++/CLI

```
void InsertFeatureShellAddThickness(  
&    System.double Thickness  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness
Shell thickness in meters

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertFeatureShellAddThickness](#).

Remarks

This method works with [IModelDoc2::InsertFeatureShell](#) to create a multi-thickness shell as follows:

1. Select the faces to remove to create the shell feature by calling [IModelDocExtension::SelectByID2](#) with Marks of 1.
2. Select faces with alternate thicknesses using Marks of 2.
3. Call this method once for each of the faces selected in step 2.
4. Call IModelDoc2::InsertFeatureShell.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertFramePoint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertFramePoint Method (IModelDoc2)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertFramePoint( _  
    ByVal Xx As System.Double, _  
    ByVal Yy As System.Double, _  
    ByVal Zz As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Xx As System.Double  
Dim Yy As System.Double  
Dim Zz As System.Double  
  
instance.InsertFramePoint(Xx, Yy, Zz)
```

C#

```
void InsertFramePoint(  
    System.double Xx,
```

```
System.double Yy,  
System.double Zz  
)
```

C++/CLI

```
void InsertFramePoint(  
& System.double Xx,  
& System.double Yy,  
& System.double Zz  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xx

Yy

Zz

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertFramePoint](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertGtol Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertGtol Method (IModelDoc2)

Creates a new geometric tolerance symbol (GTol) in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertGtol() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Object  
  
value = instance.InsertGtol()
```

C#

```
System.object InsertGtol()
```

C++/CLI

```
System.Object^ InsertGtol();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [GTol](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertGtol](#).

Example

[Insert GTol \(C#\)](#)
[Insert GTol \(VB.NET\)](#)
[Insert GTol \(VBA\)](#)

Remarks

The leader attachment point for the newly created GTol object comes from the selection made before calling this method. The initial location of the symbol is near the selection location. If there is no selection, then the GTol does not have a leader, is free standing, and is initially at the origin of the model or drawing.

This method creates an empty symbol. To fill in the text and symbols of this GTol, use the pointer returned by this method to access the various get and set methods of the [IGTol](#) interface, such as [IGTol::SetFrameSymbols2](#) and [IGTol::SetFrameValues](#). Use [IGTol::GetAnnotation](#) to retrieve the [IAnnotation](#) object.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertGtol Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertHatchedFace Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertHatchedFace Method (IModelDoc2)

Hatches the selected faces or closed sketch segments in a drawing.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertHatchedFace()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertHatchedFace()
```

C#

```
void InsertHatchedFace()
```

C++/CLI

```
void InsertHatchedFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertHatchedFace](#).

Example

[Insert Hatch \(VBA\)](#)
[Get Area Hatch Data \(VBA\)](#)
[Get Area Hatch Data \(VB.NET\)](#)
[Get Area Hatch Data \(C#\)](#)

Remarks

This method supports hatching of drawings only; it does not support hatching of parts or assemblies.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IView::GetFaceHatchCount Method](#)
[IView::GetFaceHatches Method](#)
[IView::IGetFaceHatches Method](#)
[IView::ScaleHatchPattern Property](#)
[IFaceHatch Interface](#)
[ISketchManager::CreateRegionHatch Method](#)
[ISketchManager::CreateBoundaryHatch Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertHelix Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertHelix Method (IModelDoc2)

Creates a constant-pitch helix or spiral.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertHelix( _  
    ByVal Reversed As System.Boolean, _  
    ByVal Clockwised As System.Boolean, _  
    ByVal Tapered As System.Boolean, _  
    ByVal Outward As System.Boolean, _  
    ByVal Helixdef As System.Integer, _  
    ByVal Height As System.Double, _  
    ByVal Pitch As System.Double, _  
    ByVal Revolution As System.Double, _  
    ByVal TaperAngle As System.Double, _  
    ByVal Startangle As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Reversed As System.Boolean  
Dim Clockwised As System.Boolean  
Dim Tapered As System.Boolean  
Dim Outward As System.Boolean  
Dim Helixdef As System.Integer  
Dim Height As System.Double  
Dim Pitch As System.Double  
Dim Revolution As System.Double  
Dim TaperAngle As System.Double  
Dim Startangle As System.Double
```

```
instance.InsertHelix(Reversed, Clockwised, Tapered, Outward, Helixdef, Height, Pitch, Revolution,  
TaperAngle, Startangle)
```

C#

```
void InsertHelix(  
    System.bool Reversed,  
    System.bool Clockwised,  
    System.bool Tapered,  
    System.bool Outward,  
    System.int Helixdef,  
    System.double Height,  
    System.double Pitch,  
    System.double Revolution,  
    System.double TaperAngle,
```

```
        System.double Startangle
    )
}
```

C++/CLI

```
void InsertHelix(
& System.bool Reversed,
& System.bool Clockwised,
& System.bool Tapered,
& System.bool Outward,
& System.int Helixdef,
& System.double Height,
& System.double Pitch,
& System.double Revolution,
& System.double TaperAngle,
& System.double Startangle
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Reversed

True creates a constant-pitch helix or spiral in the opposite direction of the circle used to define the diameter, false creates a constant-pitch helix or spiral in the direction of the circle's normal vector; the normal vector of a circle, for example, would be off the screen if the circle were drawn in a counter-clockwise direction

Clockwised

True for clockwise, false for counter-clockwise

Tapered

True to taper the constant-pitch helix, false for no taper

Outward

True to taper the constant-pitch helix outward, false for inward

Helixdef

Constant-pitch helix or spiral definition as defined by [swHelixDefinedBy_e](#)

Height

Height of constant-pitch helix in meters

Pitch

Constant-pitch helix or spiral pitch

Revolution

Number of revolutions

TaperAngle

Taper angle of constant-pitch helix

Startangle

Start angle rotation from base axis in radians

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertHelix](#).

Example

[Create Spiral \(C#\)](#)

[Create Spiral \(VB.NET\)](#)

[Create Spiral \(VBA\)](#)

Remarks

When creating a spiral, you must specify the pitch and number of revolutions.

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IHelixFeatureData Interface](#)

[IFeatureManager::InsertVariablePitchHelix Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertLibraryFeature Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertLibraryFeature Method (IModelDoc2)

Obsolete. See **Remarks**.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertLibraryFeature( _  
    ByVal LibFeatPartNameIn As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim LibFeatPartNameIn As System.String  
  
instance.InsertLibraryFeature(LibFeatPartNameIn)
```

C#

```
void InsertLibraryFeature(  
    System.string LibFeatPartNameIn  
)
```

C++/CLI

```
void InsertLibraryFeature(  
& System.String^ LibFeatPartNameIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

LibFeatPartNameIn
Name of the library feature

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertLibraryFeature](#).

Remarks

See [Library Features and LibraryFeatureData Objects](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ILibraryFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertLoftRefSurface Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertLoftRefSurface Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::InsertLoftRefSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertLoftRefSurface( _  
    ByVal Closed As System.Boolean, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Closed As System.Boolean  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
  
instance.InsertLoftRefSurface(Closed, KeepTangency, ForceNonRational)
```

C#

```
void InsertLoftRefSurface(  
    System.bool Closed,  
    System.bool KeepTangency,  
    System.bool ForceNonRational  
)
```

C++/CLI

```
void InsertLoftRefSurface(  
& System.bool Closed,  
& System.bool KeepTangency,  
& System.bool ForceNonRational  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

KeepTangency

ForceNonRational

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertLoftRefSurface](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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InsertLoftRefSurface2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertLoftRefSurface2 Method (IModelDoc2)

Creates a loft surface from the selected profiles, centerline, and guide curves.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertLoftRefSurface2( _  
    ByVal Closed As System.Boolean, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean, _  
    ByVal TessToleranceFactor As System.Double, _  
    ByVal StartMatchingType As System.Short, _
```

```
ByVal EndMatchingType As System.Short _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Closed As System.Boolean  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
Dim TessToleranceFactor As System.Double  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short
```

```
instance.InsertLoftRefSurface2(Closed, KeepTangency, ForceNonRational, TessToleranceFactor,  
StartMatchingType, EndMatchingType)
```

C#

```
void InsertLoftRefSurface2(  
    System.bool Closed,  
    System.bool KeepTangency,  
    System.bool ForceNonRational,  
    System.double TessToleranceFactor,  
    System.short StartMatchingType,  
    System.short EndMatchingType  
)
```

C++/CLI

```
void InsertLoftRefSurface2(  
& System.bool Closed,  
& System.bool KeepTangency,  
& System.bool ForceNonRational,  
& System.double TessToleranceFactor,  
& System.short StartMatchingType,  
& System.short EndMatchingType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

True for closed loft, false for open loft; if True, then you must have at least three profiles selected, and if you are using guide curves, the guide curves must be closed

KeepTangency

If the section curves are tangent, then you have the option to specify whether the resulting surfaces are also be tangent; specify True to maintain the tangency as seen in the section curves, false otherwise; when generating tangent surfaces, SOLIDWORKS maintains planar and cylindrical surface shapes if the section curves exhibit these characteristics

ForceNonRational

True to force the resulting surface to be non-rational, false to not

TessToleranceFactor

Factor to control the number of intermediate sections used for loft with centerline; default value is 1.0; the greater the variable, the more intermediate sections created

StartMatchingType

Tangency type at the start profile (see **Remarks**)

EndMatchingType

Tangency type at the end profile (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertLoftRefSurface2](#).

Remarks

Selection of guide curves and centerline is optional; however, selection of the profiles must be in an order consistent with the desired direction of the loft. Because you are creating a surface, the section profiles can be open.

Use of guide curves is strongly recommended, especially when selection of profiles is done in the FeatureManager design tree.

You can use any number of profiles; however, if you have selected only one profile, then any selected guide curves must be closed curves.

Use [IModelDocExtension::SelectByID2](#) to select the profiles and guide curves. The mark for:

- profile selections should be 1
- any guide curve selection, if provided, should be 2

- centerline selection, if provided, should be 4
 - start tangency vector selection, if provided, should be an 8
 - start tangency faces selection, if provided, should be a 16 (not available currently)
 - end tangency vector selection, if provided, should be a 32
 - end tangency faces selection, if provided, should be a 64 (not available currently)
- Linear edge, sketch line, axis, plane and planar faces are qualified for tangency vector sections.

The tangency types can be one of the following:

0 =	none
1 =	tangent to the normal of the profile
2 =	tangent to a selected vector
3 =	tangency to all the adjacent faces sharing an edge with the start profile
4 =	tangent to some of the selected faces sharing an edge with the start profile (not available at this moment).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IFeatureManager::InsertProtrusionBlend Method](#)
[IFeatureManager::InsertCutBlend Method](#)
[ILoftFeatureData Interface](#)
[IModeler::CreateLoftSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertMacroFeature Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertMacroFeature Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertMacroFeature3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMacroFeature( _
    ByVal CmdFile As System.String, _
    ByVal CmdModule As System.String, _
    ByVal CmdProcedure As System.String, _
    ByVal ParamNames As System.Object, _
    ByVal ParamTypes As System.Object, _
    ByVal ParamValues As System.Object, _
    ByVal PmFile As System.String, _
    ByVal PmModule As System.String, _
    ByVal PmProcedure As System.String _
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim CmdFile As System.String
Dim CmdModule As System.String
Dim CmdProcedure As System.String
Dim ParamNames As System.Object
Dim ParamTypes As System.Object
Dim ParamValues As System.Object
Dim PmFile As System.String
Dim PmModule As System.String
```



```
Dim PmProcedure As System.String
Dim value As System.Object
```

```
value = instance.InsertMacroFeature(CmdFile, CmdModule, CmdProcedure, ParamNames, ParamTypes,
ParamValues, PmFile, PmModule, PmProcedure)
```

C#

```
System.Object InsertMacroFeature(
    System.String CmdFile,
    System.String CmdModule,
    System.String CmdProcedure,
    System.Object ParamNames,
    System.Object ParamTypes,
    System.Object ParamValues,
    System.String PmFile,
    System.String PmModule,
    System.String PmProcedure
)
```

C++/CLI

```
System.Object^ InsertMacroFeature(
    & System.String^ CmdFile,
    & System.String^ CmdModule,
    & System.String^ CmdProcedure,
    & System.Object^ ParamNames,
    & System.Object^ ParamTypes,
    & System.Object^ ParamValues,
    & System.String^ PmFile,
    & System.String^ PmModule,
    & System.String^ PmProcedure
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CmdFile

CmdModule

CmdProcedure

ParamNames

ParamTypes

ParamValues

PmFile

PmModule

PmProcedure

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertMacroFeature](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertMfDraft Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertMfDraft Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertMultifaceDraft](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertMfDraft( _  
    ByVal Angle As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal IsEdgeDraft As System.Boolean, _  
    ByVal PropType As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Angle As System.Double  
Dim FlipDir As System.Boolean  
Dim IsEdgeDraft As System.Boolean  
Dim PropType As System.Integer  
  
instance.InsertMfDraft(Angle, FlipDir, IsEdgeDraft, PropType)
```

C#

```
void InsertMfDraft(  
    System.double Angle,  
    System.bool FlipDir,  
    System.bool IsEdgeDraft,  
    System.int PropType  
)
```

C++/CLI

```
void InsertMfDraft(  
& System.double Angle,  
& System.bool FlipDir,  
& System.bool IsEdgeDraft,  
& System.int PropType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

FlipDir

IsEdgeDraft

PropType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertMfDraft](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertMfDraft2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertMfDraft2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertMultifaceDraft](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertMfDraft2( _  
    ByVal Angle As System.Double, _
```

```

ByVal FlipDir As System.Boolean, _
ByVal IsEdgeDraft As System.Boolean, _
ByVal PropType As System.Integer, _
ByVal StepDraft As System.Boolean _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Angle As System.Double
Dim FlipDir As System.Boolean
Dim IsEdgeDraft As System.Boolean
Dim PropType As System.Integer
Dim StepDraft As System.Boolean

instance.InsertMfDraft2(Angle, FlipDir, IsEdgeDraft, PropType, StepDraft)

```

C#

```

void InsertMfDraft2(
    System.double Angle,
    System.bool FlipDir,
    System.bool IsEdgeDraft,
    System.int PropType,
    System.bool StepDraft
)

```

C++/CLI

```

void InsertMfDraft2(
    & System.double Angle,
    & System.bool FlipDir,
    & System.bool IsEdgeDraft,
    & System.int PropType,
    & System.bool StepDraft
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

FlipDir

IsEdgeDraft

PropType

StepDraft

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertMfDraft2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMidSurfaceExt Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertMidSurfaceExt Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertMidSurface](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMidSurfaceExt( _  
    ByVal Placement As System.Double, _  
    ByVal KnitFlag As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Placement As System.Double  
Dim KnitFlag As System.Boolean  
Dim value As System.Object
```

```
value = instance.InsertMidSurfaceExt(Placement, KnitFlag)
```

C#

```
System.object InsertMidSurfaceExt(  
    System.double Placement,  
    System.bool KnitFlag  
)
```

C++/CLI

```
System.Object^ InsertMidSurfaceExt(  
& System.double Placement,  
& System.bool KnitFlag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Placement

Value from -1 to 1 that indicates the desired location of the midsurface relative to the face pair; a value of 0.0 places the midsurface equally between the face pair

KnitFlag

True to sew all the reference surfaces (neutral sheets) into a single sheet(surface) body, false for the midsurface feature to contain individual reference surface geometry

Return Value

[Mid-surface feature](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertMidSurfaceExt](#).

Remarks

A midsurface feature is calculated from the solid body in your part document. This method first finds all parallel pairs of faces from the part solid. For each pair of parallel faces found, it creates a neutral reference surface between the two faces. The placement position of the reference surface is controlled by the placement parameter. This process repeats itself until all parallel body faces are processed. At the end of this process, all the reference surfaces are sewn into a single reference surface if knitFlag is set to True.

The hierarchy of a midsurface feature is:

A midsurface feature contains one or more reference surfaces. If sewn successfully using the knitFlag = True option, then the midsurface feature contains only one reference surface. Each reference surface is considered a sheet body. The sheet body has the normal topology that you would expect to find on a body object (for example, faces, edges, and so on). See the [IMidSurface2](#) object for methods that provide access to this topology.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertMidSurfaceExt Method](#)

Availability

InsertNewNote3 Method (IModelDoc2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertNewNote3 Method (IModelDoc2)

Creates a new note.

.NET Syntax

Visual Basic (Declaration)

```

Sub InsertNewNote3( _
    ByVal UpperText As System.String, _
    ByVal NoLeader As System.Boolean, _
    ByVal BentLeader As System.Boolean, _
    ByVal ArrowStyle As System.Short, _
    ByVal LeaderSide As System.Short, _
    ByVal Angle As System.Double, _
    ByVal BalloonStyle As System.Short, _
    ByVal BalloonFit As System.Short, _
    ByVal SmartArrow As System.Boolean _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim UpperText As System.String
Dim NoLeader As System.Boolean
Dim BentLeader As System.Boolean
Dim ArrowStyle As System.Short
Dim LeaderSide As System.Short
Dim Angle As System.Double
Dim BalloonStyle As System.Short
Dim BalloonFit As System.Short
Dim SmartArrow As System.Boolean

instance.InsertNewNote3(UpperText, NoLeader, BentLeader, ArrowStyle, LeaderSide, Angle,
BalloonStyle, BalloonFit, SmartArrow)

```

C#

```

void InsertNewNote3(
    System.string UpperText,
    System.bool NoLeader,
    System.bool BentLeader,
    System.short ArrowStyle,
    System.short LeaderSide,
    System.double Angle,
    System.short BalloonStyle,
    System.short BalloonFit,
    System.bool SmartArrow
)

```

C++/CLI

```

void InsertNewNote3(
    & System.String^ UpperText,
    & System.bool NoLeader,
    & System.bool BentLeader,
    & System.short ArrowStyle,
    & System.short LeaderSide,
    & System.double Angle,
    & System.short BalloonStyle,
    & System.short BalloonFit,
    & System.bool SmartArrow
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UpperText

Upper-text string to be put in the note

NoLeader

True for no leaderline, false if not

BentLeader

True for a bent leaderline, false if not

ArrowStyle

Arrowhead type as defined in [swArrowStyle_e](#)

LeaderSide

Leaderline side as defined in [swLeaderSide_e](#)

Angle

Text angle

BalloonStyle

Balloon style type as defined in [swBalloonStyle_e](#)

BalloonFit

Balloon fit type as defined in [swBalloonFit_e](#)

SmartArrow

If true then the arrow style specified in **Options > Detailing** is used for the arrows, if false then the ArrowStyle argument is used

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertNewNote3](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::InsertNote Method](#)

[IModelDoc2::InsertNote Method](#)

[INote Interface](#)

[IDrawingDoc::InsertNewNote2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertNote Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertNote Method (IModelDoc2)

Inserts a note in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertNote( _  
    ByVal Text As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Text As System.String  
Dim value As System.Object  
  
value = instance.InsertNote(Text)
```

C#

```
System.object InsertNote(  
    System.string Text  
)
```

C++/CLI

```
System.Object^ InsertNote(  
& System.String^ Text  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Text
Text string or symbol for the note (see **Remarks**)

Return Value

Newly created [note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertNote](#).

Example

[Insert Note \(VBA\)](#)
[Create Block Definition and Insert Block Instance \(VBA\)](#)
[Create Block Definition and Insert Block Instance \(C#\)](#)
[Create Block Definition and Insert Block Instance \(VB.NET\)](#)

Remarks

If you use a symbol in *Text*, it must be formatted as:

<*LibraryName-SymbolName*>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

The leader attachment points for the new note come from the selections made before calling this method. The initial location of the note will be near the selection location. If there are no selections, the note:

- Does not have a leader
- Is free standing
- Is at the origin of the model or drawing

This method creates a default note. To adjust the display characteristics of this note, you should use the pointer that is returned by this method to access the properties and get and set methods of the Note object, such as [INote::SetBalloon](#), [INote::Angle](#), and [INote::LockPosition](#). Use [INote::GetAnnotation](#) or [INote::!GetAnnotation](#) to retrieve the [Annotation](#) object.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[INote Interface](#)
[IModelDoc2::InsertNewNote3 Method](#)
[IModelDoc2::InsertNote Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertObject Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertObject Method (IModelDoc2)

Activates the Microsoft Insert Object dialog.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertObject()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.InsertObject()
```

C#

```
void InsertObject()
```

C++/CLI

```
void InsertObject();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertObject](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDocExtension::InsertObjectFromFile Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertObjectFromFile Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertObjectFromFile Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::InsertObjectFromFile](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertObjectFromFile( _  
    ByVal FilePath As System.String, _  
    ByVal CreateLink As System.Boolean, _  
    ByVal Xx As System.Double, _  
    ByVal Yy As System.Double, _  
    ByVal Zz As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
Dim FilePath As System.String
```

```
Dim CreateLink As System.Boolean
```

```
Dim Xx As System.Double
```

```
Dim Yy As System.Double
```

```
Dim Zz As System.Double
```

```
Dim value As System.Boolean
```

```
value = instance.InsertObjectFromFile(FilePath, CreateLink, Xx, Yy, Zz)
```

C#

```
System.bool InsertObjectFromFile(  
    System.string FilePath,  
    System.bool CreateLink,  
    System.double Xx,  
    System.double Yy,  
    System.double Zz  
)
```

C++/CLI

```
System.bool InsertObjectFromFile(  
& System.String^ FilePath,  
& System.bool CreateLink,  
& System.double Xx,  
& System.double Yy,  
& System.double Zz  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePath

CreateLink

Xx

Yy

Zz

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertObjectFromFile](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertOffsetSurface Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertOffsetSurface Method (IModelDoc2)

Inserts an offset surface.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertOffsetSurface( _  
    ByVal Thickness As System.Double, _  
    ByVal Reverse As System.Boolean _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Thickness As System.Double  
Dim Reverse As System.Boolean
```

```
instance.InsertOffsetSurface(Thickness, Reverse)
```

C#

```
void InsertOffsetSurface(  
    System.double Thickness,
```



```
        System.bool Reverse
    )
```

C++/CLI

```
void InsertOffsetSurface(
&    System.double Thickness,
&    System.bool Reverse
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness

Offset of surface from reference

Reverse

True to reverse the direction of the offset, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertOffsetSurface](#).

Remarks

Make the selections using [IModelDocExtension::SelectByID2](#) before calling this method.

See the SOLIDWORKS Help for more information about what entities are valid for selection.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISurfaceOffsetFeatureData Interface](#)
[IBody2::CreateOffsetSurface Method](#)
[IBody2::ICreateOffsetSurface Method](#)
[IBody2::MakeOffset Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[ISurface::GetOffsetSurfParams2 Method](#)
[ISurface::IGetOffsetSurfParams2 Method](#)
[ISurface::IsOffset Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertPlanarRefSurface Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertPlanarRefSurface Method (IModelDoc2)

Inserts a planar reference surface.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertPlanarRefSurface() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean
```

```
value = instance.InsertPlanarRefSurface()
```

C#

```
System.bool InsertPlanarRefSurface()
```

C++/CLI

```
System.bool InsertPlanarRefSurface();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the reference surface is created, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertPlanarRefSurface](#).

Example

[Create Planar Surface Feature \(VBA\)](#)
[Create Planar Surface Feature \(VB.NET\)](#)
[Create Planar Surface Feature \(C#\)](#)

Remarks

Before calling this method, select the surface's boundary entities using [IModelDocExtension::SelectById2](#) with Marks of 1. See the SOLIDWORKS Help for more information about what entities are valid for selection.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISurfacePlanarFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertPoint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertPoint Method (IModelDoc2)

Inserts a point in this model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertPoint()
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
instance.InsertPoint()
```

C#

```
void InsertPoint()
```

C++/CLI

```
void InsertPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertPoint](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertProjectedSketch Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProjectedSketch Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::InsertProjectedSketch2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertProjectedSketch()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertProjectedSketch()
```

C#

```
void InsertProjectedSketch()
```

C++/CLI

```
void InsertProjectedSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProjectedSketch](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertProjectedSketch2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProjectedSketch2 Method (IModelDoc2)

Obsolete. See [IProjectionCurveFeatureData](#) and [IFeatureManager::CreateDefinition](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertProjectedSketch2( _  
    ByVal Reverse As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Reverse As System.Integer
```

```
Dim value As System.Object  
  
value = instance.InsertProjectedSketch2(Reverse)
```

C#

```
System.object InsertProjectedSketch2(  
    System.int Reverse  
)
```

C++/CLI

```
System.Object^ InsertProjectedSketch2(  
& System.int Reverse  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Reverse

1 to reverse the projected direction, 0 to not reverse the projected direction

Return Value

Newly created [feature](#) or Nothing or null if the operation fails

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProjectedSketch2](#).

Example

[Get Projected Curve Feature Data \(C#\)](#)
[Get Projected Curve Feature Data \(VB.NET\)](#)
[Get Projected Curve Feature Data \(VBA\)](#)

Remarks

You can reverse the direction in which the curve is projected. This is necessary only when the selected face wraps around the plane of the curve. For example, if the sketch being projected is surrounded by a cylindrical face, then two possible projections exist. The reverse argument switches the direction based on the normal vector of the sketch. The default direction is along the sketch normal.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertProjectedSketch2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertProtrusionBlend Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProtrusionBlend Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionBlend](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertProtrusionBlend( _  
    ByVal Closed As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Closed As System.Boolean

instance.InsertProtrusionBlend(Closed)
```

C#

```
void InsertProtrusionBlend(
    System.bool Closed
)
```

C++/CLI

```
void InsertProtrusionBlend(
    & System.bool Closed
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProtrusionBlend](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertProtrusionBlend2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProtrusionBlend2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionBlend](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertProtrusionBlend2( _
    ByVal Closed As System.Boolean, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Closed As System.Boolean
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
```

```
instance.InsertProtrusionBlend2(Closed, KeepTangency, ForceNonRational)
```

C#

```
void InsertProtrusionBlend2(
    System.bool Closed,
    System.bool KeepTangency,
    System.bool ForceNonRational
)
```

C++/CLI

```
void InsertProtrusionBlend2(
& System.bool Closed,
& System.bool KeepTangency,
& System.bool ForceNonRational
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

KeepTangency

ForceNonRational

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProtrusionBlend2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertProtrusionBlend3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProtrusionBlend3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionBlend](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertProtrusionBlend3( _
    ByVal Closed As System.Boolean, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal TessToleranceFactor As System.Double, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Closed As System.Boolean
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim TessToleranceFactor As System.Double
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
```

```
instance.InsertProtrusionBlend3(Closed, KeepTangency, ForceNonRational, TessToleranceFactor,
StartMatchingType, EndMatchingType)
```

C#

```
void InsertProtrusionBlend3(
    System.bool Closed,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.double TessToleranceFactor,
    System.short StartMatchingType,
    System.short EndMatchingType
)
```

```
void InsertProtrusionBlend3(
& System.bool Closed,
& System.bool KeepTangency,
& System.bool ForceNonRational,
& System.double TessToleranceFactor,
& System.short StartMatchingType,
& System.short EndMatchingType
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

KeepTangency

ForceNonRational

TessToleranceFactor

StartMatchingType

EndMatchingType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProtrusionBlend3](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertProtrusionBlend4 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProtrusionBlend4 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionBlend](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertProtrusionBlend4( _
    ByVal Closed As System.Boolean, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal TessToleranceFactor As System.Double, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short, _
    ByVal IsThinBody As System.Boolean, _
    ByVal Thickness1 As System.Double, _
    ByVal Thickness2 As System.Double, _
    ByVal ThinType As System.Short _
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim Closed As System.Boolean
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim TessToleranceFactor As System.Double
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
```



```
Dim Thickness2 As System.Double
Dim ThinType As System.Short
```

```
instance.InsertProtrusionBlend4(Closed, KeepTangency, ForceNonRational, TessToleranceFactor,
StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2, ThinType)
```

C#

```
void InsertProtrusionBlend4(
    System.bool Closed,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.double TessToleranceFactor,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType
)
```

C++/CLI

```
void InsertProtrusionBlend4(
& System.bool Closed,
& System.bool KeepTangency,
& System.bool ForceNonRational,
& System.double TessToleranceFactor,
& System.short StartMatchingType,
& System.short EndMatchingType,
& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.short ThinType
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

KeepTangency

ForceNonRational

TessToleranceFactor

StartMatchingType

EndMatchingType

IsThinBody

Thickness1

Thickness2

ThinType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProtrusionBlend4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertProtrusionSwept Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProtrusionSwept Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertProtrusionSwept( _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal KeepNormalConstant As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Propagate As System.Boolean  
Dim Alignment As System.Boolean  
Dim KeepNormalConstant As System.Boolean  
  
instance.InsertProtrusionSwept(Propagate, Alignment, KeepNormalConstant)
```

C#

```
void InsertProtrusionSwept(  
    System.bool Propagate,  
    System.bool Alignment,  
    System.bool KeepNormalConstant  
)
```

C++/CLI

```
void InsertProtrusionSwept(  
& System.bool Propagate,  
& System.bool Alignment,  
& System.bool KeepNormalConstant  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

KeepNormalConstant

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProtrusionSwept](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertProtrusionSwept2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProtrusionSwept2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager.InsertProtrusionSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertProtrusionSwept2( _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal TwistCtrlOption As System.Short, _
```

```
ByVal KeepTangency As System.Boolean, _  
ByVal ForceNonRational As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Propagate As System.Boolean  
Dim Alignment As System.Boolean  
Dim TwistCtrlOption As System.Short  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean
```

```
instance.InsertProtrusionSwept2(Propagate, Alignment, TwistCtrlOption, KeepTangency,  
ForceNonRational)
```

C#

```
void InsertProtrusionSwept2(  
    System.bool Propagate,  
    System.bool Alignment,  
    System.short TwistCtrlOption,  
    System.bool KeepTangency,  
    System.bool ForceNonRational  
)
```

C++/CLI

```
void InsertProtrusionSwept2(  
& System.bool Propagate,  
& System.bool Alignment,  
& System.short TwistCtrlOption,  
& System.bool KeepTangency,  
& System.bool ForceNonRational  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProtrusionSwept2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertProtrusionSwept3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProtrusionSwept3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub InsertProtrusionSwept3( _
    ByVal Propagate As System.Boolean, _
    ByVal Alignment As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short

instance.InsertProtrusionSwept3(Propagate, Alignment, TwistCtrlOption, KeepTangency,
ForceNonRational, StartMatchingType, EndMatchingType)

```

C#

```

void InsertProtrusionSwept3(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.short StartMatchingType,
    System.short EndMatchingType
)

```

C++/CLI

```

void InsertProtrusionSwept3(
    & System.bool Propagate,
    & System.bool Alignment,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.short StartMatchingType,
    & System.short EndMatchingType
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProtrusionSwept3](#).

See Also

InsertProtrusionSwept4 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertProtrusionSwept4 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertProtrusionSwept4( _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal TwistCtrlOption As System.Short, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Short _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Propagate As System.Boolean  
Dim Alignment As System.Boolean  
Dim TwistCtrlOption As System.Short  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short  
Dim IsThinBody As System.Boolean  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ThinType As System.Short  
  
instance.InsertProtrusionSwept4(Propagate, Alignment, TwistCtrlOption, KeepTangency,  
ForceNonRational, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,  
ThinType)
```

C#

```
void InsertProtrusionSwept4(  
    System.bool Propagate,  
    System.bool Alignment,  
    System.short TwistCtrlOption,  
    System.bool KeepTangency,  
    System.bool ForceNonRational,  
    System.short StartMatchingType,  
    System.short EndMatchingType,  
    System.bool IsThinBody,  
    System.double Thickness1,  
    System.double Thickness2,  
    System.short ThinType  
)
```

C++/CLI

```
void InsertProtrusionSwept4(  
& System.bool Propagate,  
& System.bool Alignment,  
& System.short TwistCtrlOption,
```

```
& System.bool KeepTangency,  
& System.bool ForceNonRational,  
& System.short StartMatchingType,  
& System.short EndMatchingType,  
& System.bool IsThinBody,  
& System.double Thickness1,  
& System.double Thickness2,  
& System.short ThinType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

IsThinBody

Thickness1

Thickness2

ThinType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProtrusionSwept4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertRadiateSurface Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertRadiateSurface Method (IModelDoc2)

Creates a radiate surface based on the selections.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertRadiateSurface(  
    ByVal Distance As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal TangentPropagate As System.Boolean _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Distance As System.Double  
Dim FlipDir As System.Boolean  
Dim TangentPropagate As System.Boolean  
  
instance.InsertRadiateSurface(Distance, FlipDir, TangentPropagate)
```

C#

```
void InsertRadiateSurface(  
    System.double Distance,
```



```
System.bool FlipDir,  
System.bool TangentPropagate  
)
```

C++/CLI

```
void InsertRadiateSurface(  
& System.double Distance,  
& System.bool FlipDir,  
& System.bool TangentPropagate  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Distance

Distance to extend the surface

FlipDir

True to flip the direction; by default the direction is out from the center of the face

TangentPropagate

True to propagate the surface along tangent faces, false limits the surface to the selected face

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertRadiateSurface](#).

Example

[Create Radiate Surface Feature \(VBA\)](#)
[Create Radiate Surface Feature \(VB.NET\)](#)
[Create Radiate Surface Feature \(C#\)](#)

Remarks

Before calling this method, call [IModelDocExtension::SelectById2](#) to select

- Reference direction using Mark = 1.
- Radiate entities using Mark = 2.

See the SOLIDWORKS Help for information about what entities are valid for selection.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISurfaceRadiateFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertRefPoint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertRefPoint Method (IModelDoc2)

Inserts a reference point based on the current selections.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertRefPoint() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim value As System.Boolean
```



```
value = instance.InsertRefPoint()
```

C#

```
System.bool InsertRefPoint()
```

C++/CLI

```
System.bool InsertRefPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertRefPoint](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IRefPoint Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertRevolvedRefSurface Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertRevolvedRefSurface Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRevolvedRefSurface](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertRevolvedRefSurface( _  
    ByVal Angle As System.Double, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal Angle2 As System.Double, _  
    ByVal RevType As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Angle As System.Double  
Dim ReverseDir As System.Boolean  
Dim Angle2 As System.Double  
Dim RevType As System.Integer
```

```
instance.InsertRevolvedRefSurface(Angle, ReverseDir, Angle2, RevType)
```

C#

```
void InsertRevolvedRefSurface(  
    System.double Angle,  
    System.bool ReverseDir,  
    System.double Angle2,  
    System.int RevType  
)
```

C++/CLI

```
void InsertRevolvedRefSurface(
& System.double Angle,
& System.bool ReverseDir,
& System.double Angle2,
& System.int RevType
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

ReverseDir

Angle2

RevType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertRevolvedRefSurface](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertRib Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertRib Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::InsertRib2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertRib( _
    ByVal Is2Sided As System.Boolean, _
    ByVal ReverseThicknessDir As System.Boolean, _
    ByVal Thickness As System.Double, _
    ByVal ReferenceEdgeIndex As System.Integer, _
    ByVal ReverseMaterialDir As System.Boolean, _
    ByVal IsDrafted As System.Boolean, _
    ByVal DraftOutward As System.Boolean, _
    ByVal DraftAngle As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Is2Sided As System.Boolean
Dim ReverseThicknessDir As System.Boolean
Dim Thickness As System.Double
Dim ReferenceEdgeIndex As System.Integer
Dim ReverseMaterialDir As System.Boolean
Dim IsDrafted As System.Boolean
Dim DraftOutward As System.Boolean
Dim DraftAngle As System.Double
```

```
instance.InsertRib(Is2Sided, ReverseThicknessDir, Thickness, ReferenceEdgeIndex,
ReverseMaterialDir, IsDrafted, DraftOutward, DraftAngle)
```

C#

```
void InsertRib(
    System.bool Is2Sided,
```

```
System.bool ReverseThicknessDir,  
System.double Thickness,  
System.int ReferenceEdgeIndex,  
System.bool ReverseMaterialDir,  
System.bool IsDrafted,  
System.bool DraftOutward,  
System.double DraftAngle  
)
```

C++/CLI

```
void InsertRib(  
& System.bool Is2Sided,  
& System.bool ReverseThicknessDir,  
& System.double Thickness,  
& System.int ReferenceEdgeIndex,  
& System.bool ReverseMaterialDir,  
& System.bool IsDrafted,  
& System.bool DraftOutward,  
& System.double DraftAngle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Is2Sided

ReverseThicknessDir

Thickness

ReferenceEdgeIndex

ReverseMaterialDir

IsDrafted

DraftOutward

DraftAngle

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertRib](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertRib2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertRib2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRib](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertRib2( _  
    ByVal Is2Sided As System.Boolean, _  
    ByVal ReverseThicknessDir As System.Boolean, _  
    ByVal Thickness As System.Double, _  
    ByVal ReferenceEdgeIndex As System.Integer, _  
    ByVal ReverseMaterialDir As System.Boolean, _  
    ByVal IsDrafted As System.Boolean, _  
    ByVal DraftOutward As System.Boolean, _  
    ByVal DraftAngle As System.Double, _  
    ByVal IsNormToSketch As System.Boolean _  
)
```

```

Dim instance As IModelDoc2
Dim Is2Sided As System.Boolean
Dim ReverseThicknessDir As System.Boolean
Dim Thickness As System.Double
Dim ReferenceEdgeIndex As System.Integer
Dim ReverseMaterialDir As System.Boolean
Dim IsDrafted As System.Boolean
Dim DraftOutward As System.Boolean
Dim DraftAngle As System.Double
Dim IsNormToSketch As System.Boolean

instance.InsertRib2(Is2Sided, ReverseThicknessDir, Thickness, ReferenceEdgeIndex,
ReverseMaterialDir, IsDrafted, DraftOutward, DraftAngle, IsNormToSketch)

```

C#

```

void InsertRib2(
    System.bool Is2Sided,
    System.bool ReverseThicknessDir,
    System.double Thickness,
    System.int ReferenceEdgeIndex,
    System.bool ReverseMaterialDir,
    System.bool IsDrafted,
    System.bool DraftOutward,
    System.double DraftAngle,
    System.bool IsNormToSketch
)

```

C++/CLI

```

void InsertRib2(
& System.bool Is2Sided,
& System.bool ReverseThicknessDir,
& System.double Thickness,
& System.int ReferenceEdgeIndex,
& System.bool ReverseMaterialDir,
& System.bool IsDrafted,
& System.bool DraftOutward,
& System.double DraftAngle,
& System.bool IsNormToSketch
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Is2Sided

True if the rib is double-sided rib, false if the rib is single sided; a double-sided rib thickens on both sides of the sketch

ReverseThicknessDir

For single-sided ribs, the direction thickened is the opposite side of the sketch normal

Thickness

Rib thickness

ReferenceEdgeIndex

Edge in the sketch to us to determine the material direction and for draft reference; when the rib is drafted, the mid point of this edge maintains the thickness value; other points on the rib may have a different thickness due to the draft

ReverseMaterialDir

Which direction the rib has material; it is on one side or the other side of the reference edge base on this flag

IsDrafted

True if the rib is drafted, false if not

DraftOutward

True to draft outwards, false to not

DraftAngle

Draft angle to apply to the rib

IsNormToSketch

True if the rib is normal to the sketch, false if the rib is parallel to the sketch

See [ModelDoc2::InsertRib2](#).

Remarks

This method supports multibody parts. You must select the input body by calling [IModelDocExtension::SelectByID2](#) and setting its Mark argument to 1 before calling this method.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertRip Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertRip Method (IModelDoc2)

Creates a rip feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertRip(  
    ByVal Gap As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Gap As System.Double  
  
instance.InsertRip(Gap)
```

C#

```
void InsertRip(  
    System.double Gap  
)
```

C++/CLI

```
void InsertRip(  
& System.double Gap  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Gap
Size of the rip gap

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertRip](#).

Example

[Access Edges on Rip Feature \(VBA\)](#)
[Access Edges on Rip Feature \(VB.NET\)](#)
[Access Edges on Rip Feature \(C#\)](#)

Remarks

The direction of the rip is determined by the Mark value used to select the edges along which to rip.

Direction	Selection Mark	IRipFeatureData::GetDirection
Current	16	0
Other	32	1
Both	64	2

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IRipFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertRoutePoint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertRoutePoint Method (IModelDoc2)

Adds a route point based on the selected point.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertRoutePoint()
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
instance.InsertRoutePoint()
```

C#

```
void InsertRoutePoint()
```

C++/CLI

```
void InsertRoutePoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertRoutePoint](#).

Remarks

If the selection set is not complete, then the **Insert Route Point** dialog is displayed.

The route point is the point on the fitting that aligns with the sketch point when the fitting is inserted.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::EditRoute Method](#)
[IAssemblyDoc::GetRouteManager Method](#)
[IAssemblyDoc::IsRouteAssembly Method](#)
[IFeatureManager::InsertConnectionPoint Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertScale Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertScale Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager.InsertScale](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub InsertScale( _  
    ByVal ScaleFactor\_x As System.Double, _  
    ByVal ScaleFactor\_y As System.Double, _  
    ByVal ScaleFactor\_z As System.Double, _  
    ByVal IsUniform As System.Boolean, _  
    ByVal ScaleType As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ScaleFactor_x As System.Double  
Dim ScaleFactor_y As System.Double  
Dim ScaleFactor_z As System.Double  
Dim IsUniform As System.Boolean  
Dim ScaleType As System.Integer  
  
instance.InsertScale(ScaleFactor_x, ScaleFactor_y, ScaleFactor_z, IsUniform, ScaleType)
```

C#

```
void InsertScale(  
    System.double ScaleFactor\_x,  
    System.double ScaleFactor\_y,  
    System.double ScaleFactor\_z,  
    System.bool IsUniform,  
    System.int ScaleType  
)
```

C++/CLI

```
void InsertScale(  
& System.double ScaleFactor\_x,  
& System.double ScaleFactor\_y,  
& System.double ScaleFactor\_z,  
& System.bool IsUniform,  
& System.int ScaleType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ScaleFactor_x

ScaleFactor_y

ScaleFactor_z

IsUniform

ScaleType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2.InsertScale](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSewRefSurface Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSewRefSurface Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertSewRefSurface](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSewRefSurface()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertSewRefSurface()
```

C#

```
void InsertSewRefSurface()
```

C++/CLI

```
void InsertSewRefSurface();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSewRefSurface](#).

Remarks

Make selections using [IModelDocExtension::SelectByID2](#) with a mark number of 1. See the SOLIDWORKS Help for information about what entities are valid for selection.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISurfaceKnitFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertSheetMetal3dBend Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetal3dBend Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetal3dBend](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSheetMetal3dBend(  
    ByVal Angle As System.Double, _
```

```

ByVal Radius As System.Double, _
ByVal FlipDir As System.Boolean, _
ByVal BendPos As System.Short _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Angle As System.Double
Dim Radius As System.Double
Dim FlipDir As System.Boolean
Dim BendPos As System.Short

instance.InsertSheetMetal3dBend(Angle, Radius, FlipDir, BendPos)

```

C#

```

void InsertSheetMetal3dBend(
    System.double Angle,
    System.double Radius,
    System.bool FlipDir,
    System.short BendPos
)

```

C++/CLI

```

void InsertSheetMetal3dBend(
    & System.double Angle,
    & System.double Radius,
    & System.bool FlipDir,
    & System.short BendPos
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

Radius

FlipDir

BendPos

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetal3dBend](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSheetMetalBaseFlange Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalBaseFlange Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalBaseFlange](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub InsertSheetMetalBaseFlange( _
    ByVal Thickness As System.Double, _
    ByVal ThickenDir As System.Boolean, _
    ByVal Radius As System.Double, _
    ByVal ExtrudeDist1 As System.Double, _

```

```

ByVal ExtrudeDist2 As System.Double, _
ByVal FlipExtruDir As System.Boolean, _
ByVal EndCondition1 As System.Integer, _
ByVal EndCondition2 As System.Integer, _
ByVal DirToUse As System.Integer _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Thickness As System.Double
Dim ThickenDir As System.Boolean
Dim Radius As System.Double
Dim ExtrudeDist1 As System.Double
Dim ExtrudeDist2 As System.Double
Dim FlipExtruDir As System.Boolean
Dim EndCondition1 As System.Integer
Dim EndCondition2 As System.Integer
Dim DirToUse As System.Integer

```

```

instance.InsertSheetMetalBaseFlange(Thickness, ThickenDir, Radius, ExtrudeDist1, ExtrudeDist2,
FlipExtruDir, EndCondition1, EndCondition2, DirToUse)

```

C#

```

void InsertSheetMetalBaseFlange(
    System.double Thickness,
    System.bool ThickenDir,
    System.double Radius,
    System.double ExtrudeDist1,
    System.double ExtrudeDist2,
    System.bool FlipExtruDir,
    System.int EndCondition1,
    System.int EndCondition2,
    System.int DirToUse
)

```

C++/CLI

```

void InsertSheetMetalBaseFlange(
& System.double Thickness,
& System.bool ThickenDir,
& System.double Radius,
& System.double ExtrudeDist1,
& System.double ExtrudeDist2,
& System.bool FlipExtruDir,
& System.int EndCondition1,
& System.int EndCondition2,
& System.int DirToUse
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness

ThickenDir

Radius

ExtrudeDist1

ExtrudeDist2

FlipExtruDir

EndCondition1

EndCondition2

DirToUse

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSheetMetalBreakCorner Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalBreakCorner Method (IModelDoc2)

Inserts a break corner into a sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSheetMetalBreakCorner( _  
    ByVal Type As System.Integer, _  
    ByVal Distance As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Type As System.Integer  
Dim Distance As System.Double  
  
instance.InsertSheetMetalBreakCorner(Type, Distance)
```

C#

```
void InsertSheetMetalBreakCorner(  
    System.int Type,  
    System.double Distance  
)
```

C++/CLI

```
void InsertSheetMetalBreakCorner(  
& System.int Type,  
& System.double Distance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Corner type as defined in [swBreakCornerTypes_e](#)

Distance

Distance to break from corner

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalBreakCorner](#).

Example

[Modify Break Corner Feature \(C#\)](#)
[Modify Break Corner Feature \(VB.NET\)](#)
[Modify Break Corner Feature \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IBreakCornerFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalClosedCorner Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalClosedCorner Method (IModelDoc2)

Inserts a sheet metal closed corner into this model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSheetMetalClosedCorner()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.InsertSheetMetalClosedCorner()
```

C#

```
void InsertSheetMetalClosedCorner()
```

C++/CLI

```
void InsertSheetMetalClosedCorner();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalClosedCorner](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IClosedCornerFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalEdgeFlange Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalEdgeFlange Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalEdgeFlange2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalEdgeFlange( _  
    ByVal FlangeEdge As System.Object, _  
    ByVal SketchFeat As System.Object, _  
    ByVal BooleanOptions As System.Integer, _  
    ByVal DAngle As System.Double, _  
    ByVal DRadius As System.Double, _  
    ByVal BendPosition As System.Integer, _  
    ByVal DOffsetDist As System.Double, _
```

```

    ByVal ReliefType As System.Integer, _
    ByVal DReliefRatio As System.Double, _
    ByVal DReliefWidth As System.Double, _
    ByVal DReliefDepth As System.Double _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim FlangeEdge As System.Object
Dim SketchFeat As System.Object
Dim BooleanOptions As System.Integer
Dim DAngle As System.Double
Dim DRadius As System.Double
Dim BendPosition As System.Integer
Dim DOffsetDist As System.Double
Dim ReliefType As System.Integer
Dim DReliefRatio As System.Double
Dim DReliefWidth As System.Double
Dim DReliefDepth As System.Double
Dim value As System.Object

```

```

value = instance.InsertSheetMetalEdgeFlange(FlangeEdge, SketchFeat, BooleanOptions, DAngle,
DRadius, BendPosition, DOffsetDist, ReliefType, DReliefRatio, DReliefWidth, DReliefDepth)

```

C#

```

System.object InsertSheetMetalEdgeFlange(
    System.object FlangeEdge,
    System.object SketchFeat,
    System.int BooleanOptions,
    System.double DAngle,
    System.double DRadius,
    System.int BendPosition,
    System.double DOffsetDist,
    System.int ReliefType,
    System.double DReliefRatio,
    System.double DReliefWidth,
    System.double DReliefDepth
)

```

C++/CLI

```

System.Object^ InsertSheetMetalEdgeFlange(
    & System.Object^ FlangeEdge,
    & System.Object^ SketchFeat,
    & System.int BooleanOptions,
    & System.double DAngle,
    & System.double DRadius,
    & System.int BendPosition,
    & System.double DOffsetDist,
    & System.int ReliefType,
    & System.double DReliefRatio,
    & System.double DReliefWidth,
    & System.double DReliefDepth
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FlangeEdge

SketchFeat

BooleanOptions

DAngle

DRadius

BendPosition

DOffsetDist

ReliefType

DReliefRatio

DReliefWidth

DReliefDepth

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalEdgeFlange](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalFold Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalFold Method (IModelDoc2)

Inserts a fold feature at the selected objects.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSheetMetalFold()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertSheetMetalFold()
```

C#

```
void InsertSheetMetalFold()
```

C++/CLI

```
void InsertSheetMetalFold();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalFold](#).

Example

[Insert and Access Fold Feature \(C#\)](#)
[Insert and Access Fold Feature \(VB.NET\)](#)
[Insert and Access Fold Feature \(VBA\)](#)

Remarks

Selects the objects for the fold feature before calling this method.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IFoldsFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0